



# Java LinkedHashMap

[< Previous](#)[Next >](#)

## Java LinkedHashMap

A `LinkedHashMap` stores keys and values, and keeps them in the same order you put them in.

It is part of the `java.util` package and implements the `Map` interface.

**Tip:** Use `LinkedHashMap` when you want predictable iteration order (insertion order).

## Create a LinkedHashMap

Create a `LinkedHashMap` object called `capitalCities` that will store `String` keys and `String` values:

## Example

[Get your own Java Server](#)



Now you can use methods like `put()` to add key/value pairs, `get()` to retrieve a value by key, and `remove()` to delete an entry - all while maintaining insertion order.

## Add Items

Use the `put()` method to add items to the `LinkedHashMap` :

### Example

```
// Import the LinkedHashMap class
import java.util.LinkedHashMap;

public class Main {
    public static void main(String[] args) {
        LinkedHashMap<String, String> capitalCities = new LinkedHashMap<>();

        capitalCities.put("England", "London");
        capitalCities.put("India", "New Dehli");
        capitalCities.put("Austria", "Wien");
        capitalCities.put("Norway", "Oslo");
        capitalCities.put("Norway", "Oslo"); // Duplicate
        capitalCities.put("USA", "Washington DC");

        System.out.println(capitalCities);
    }
}
```

Try it Yourself »



**Note:** Duplicates like "Norway" are ignored.

## Access an Item

Use `get()` with a key to get its associated value:

### Example

```
capitalCities.get("England");
```

[Try it Yourself »](#)

## Remove an Item

Use `remove()` to remove an item by key:

### Example

```
capitalCities.remove("England");
```

[Try it Yourself »](#)

Use `clear()` to remove all items:



```
capitalCities.clear();
```

[Try it Yourself »](#)

## LinkedHashMap Size

Use `size()` to check how many key/value pairs are in the map:

### Example

```
capitalCities.size();
```

[Try it Yourself »](#)

**Note:** The size only counts unique keys. If a key is added more than once, only the latest value is kept.

## Loop Through a LinkedHashMap

You can loop through a `LinkedHashMap` using a for-each loop. Use:

- `keySet()` to get all keys
- `values()` to get all values

### Example

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)[}](#)[Try it Yourself »](#)

## Example

```
// Print values
for (String value : capitalCities.values()) {
    System.out.println(value);
}
```

[Try it Yourself »](#)

## Example

```
// Print keys and values
for (String key : capitalCities.keySet()) {
    System.out.println("Key: " + key + ", Value: " + capitalCities.get(key));
}
```

[Try it Yourself »](#)

# HashMap vs LinkedHashMap

| Feature | HashMap | LinkedHashMap |
|---------|---------|---------------|
|---------|---------|---------------|



Duplicates

Keys must be unique

Keys must be unique

**Tip:** Use `LinkedHashMap` when you want the map to remember the order in which entries were added.

## The var Keyword

From Java 10, you can use the `var` keyword to declare a `LinkedHashMap` variable without writing the type twice. The compiler figures out the type from the value you assign.

This makes code shorter, **but many developers still use the full type for clarity**. Since `var` is valid Java, you may see it in other code, so it's good to know that it exists:

### Example

```
// Without var
LinkedHashMap<String, String> capitalCities = new LinkedHashMap<String, Str

// With var
var capitalCities = new LinkedHashMap<String, String>();
```

Try it Yourself »

## The Map Interface

**Note:** Sometimes you will see both `Map` and `LinkedHashMap` in Java code, like this:



```
Map<String, String> capitalCities = new LinkedHashMap<>();
```

Try it Yourself »

This means the variable (capitalCities) is declared as a **Map** (the interface), but it stores a **LinkedHashMap** object (the actual map). Since **LinkedHashMap** implements the **Map** interface, this is possible.

It works the same way, but some developers prefer this style because it gives them more flexibility to change the type later.

[< Previous](#)[Sign in to track progress](#)[Next >](#)

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)

## COLOR PICKER

[REMOVE ADS](#)[PLUS](#)[SPACES](#)[GET CERTIFIED](#)[FOR TEACHERS](#)



[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)

## Top Tutorials

[HTML Tutorial](#)  
[CSS Tutorial](#)  
[JavaScript Tutorial](#)  
[How To Tutorial](#)  
[SQL Tutorial](#)  
[Python Tutorial](#)  
[W3.CSS Tutorial](#)  
[Bootstrap Tutorial](#)  
[PHP Tutorial](#)  
[Java Tutorial](#)  
[C++ Tutorial](#)  
[jQuery Tutorial](#)

## Top References

[HTML Reference](#)  
[CSS Reference](#)  
[JavaScript Reference](#)  
[SQL Reference](#)  
[Python Reference](#)  
[W3.CSS Reference](#)  
[Bootstrap Reference](#)  
[PHP Reference](#)  
[HTML Colors](#)  
[Java Reference](#)  
[AngularJS Reference](#)  
[jQuery Reference](#)

## Top Examples

[HTML Examples](#)  
[CSS Examples](#)  
[JavaScript Examples](#)  
[How To Examples](#)  
[SQL Examples](#)  
[Python Examples](#)  
[W3.CSS Examples](#)  
[Bootstrap Examples](#)  
[PHP Examples](#)  
[Java Examples](#)  
[XML Examples](#)  
[jQuery Examples](#)

## Get Certified

[HTML Certificate](#)  
[CSS Certificate](#)  
[JavaScript Certificate](#)  
[Front End Certificate](#)  
[SQL Certificate](#)  
[Python Certificate](#)  
[PHP Certificate](#)  
[jQuery Certificate](#)  
[Java Certificate](#)  
[C++ Certificate](#)  
[C# Certificate](#)  
[XML Certificate](#)

[FORUM](#) [ABOUT](#) [ACADEMY](#)

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning.

Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full



Tutorials ▼

References ▼

Exercises ▼



Sign In



CSS

JAVASCRIPT

SQL

PYTHON

JAVA

PHP

HOW TO

W3.CSS

C